Ankush Jain

Senior Software Engineer

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SUMMARY

Senior software engineer, five years on distributed backends where correctness is not optional - service integration, event-driven architecture, cloud infrastructure, access control and applied ML. Owns systems from design to the CI/CD that ships them.

EXPERIENCE

Apra Labs

May 2023 – Present

Senior Software Engineer — May 2026 – Present
Software Engineer — May 2023 – May 2026

Engineering services across healthcare, medical devices and imaging; clients not named.

Clinical Trial Management Platform (Nov 2024 – Present)

- Built the integration hub's base adapter modules, turning every new integration from a bespoke build into a pattern
- Designed and shipped a full RBAC system: admin-only endpoints, per-role response filtering and pre-authorization checks
- Mapped business entitlements onto Microsoft Entra identity, enforcing endpoint and data-level access through JWT claims
- Remediated static-analysis security findings and led a security overhaul separating study-team and participant concerns
- Introduced server-sent events for real-time product feedback, a pattern later adopted by other teams in the organization
- Cut the backend deploy pipeline from 26m33s to 18m07s (-32%)
- Reduced a service container image from 450 MB to 290 MB (-36%) by eliminating unneeded dependencies
- Eliminated $200+/month (~$2.4k/year) of idle cloud spend by decommissioning legacy services, then added a billing review

Medicine Data Platform (May 2026 – Present)

- Modelled clinical protocols and surveys on FHIR resources with OMOP/LOINC terminology; change logs via FHIR provenance
- Drove requirements directly with the client, turning whiteboard sessions into wireframed workflows and E2E implementation

Document AI / Handwriting Recognition Pipeline (May 2025 – Nov 2025)

- Cut the deployable image from 19 GB to 5.28 GB (-72%) through packaging design
- Designed the codebase so the third-party model ships with the system and its licence swaps on expiry without a rebuild
- Reviewed code and design and guided the team, enabling them to build the LLM-based successor system

Headless Video Recording Platform (Apr 2024 – Oct 2024)

- Wrote the Node backend from scratch, managing the lifecycle of the C++ binaries and carrying inter-process communication over Redis key-value and Pub/Sub, so the whole system shared one source of truth

AR Surgical Navigation (Aug 2023 – Apr 2024)

- Cut end-to-end pipeline latency from 22s to 2s by moving the reconstruction pipeline from file-based I/O to in-memory and eliminating a per-frame signed distance field computation that dominated the frame budget
- Optimised inference by converting PyTorch models to ONNX and TensorRT, cutting per-frame model execution time
- Onboarded two junior engineers and led the project through a stretch when tech leads were unavailable

ByteLearn

May 2022 – Apr 2023

Software Engineer

- Built out monitoring and log management: Graylog for structured logging, Sentry APM across the Flask and Node.js services
- Core maintainer of the open-source mathsteps library; added complex expression support and improved parsing accuracy

Deloitte USI

Sep 2021 – May 2022

Business Technical Analyst (Software Developer)

- Extended the integration layer of an AWS Step Functions state machine, building the API calls and Lambda functions behind it
- Built the integrations each additional US state needed to move onto the new platform, which made the rollouts faster

SKILLS

Languages: Python, Java, Kotlin, TypeScript, SQL
Backend & data: Spring Boot, Node.js, REST, Server-Sent Events, RabbitMQ, GCP Pub/Sub, PostgreSQL, Redis, HL7 FHIR
Cloud & DevOps: GCP (Cloud Run), AWS (Lambda, Step Functions), Terraform, Docker, CI/CD, Observability (Sentry, Graylog)
Security & access: RBAC, Authentication & authorization, OAuth / JWT, Microsoft Entra, Static analysis remediation
ML & imaging: Computer vision, PyTorch, ONNX/TensorRT, Signal processing (FFT/DFT), OCR/HTR

EDUCATION

Dr. SPM IIIT Naya Raipur — B.Tech, Computer Science and Engineering — 2017 – 2021

ACHIEVEMENTS

Honourable Mention, ACM ICPC Kharagpur Regional Contest (2019); Honourable Mention, ACM ICPC Kolkata-Kanpur Regional Contest (2018); 3rd place, Codeutsava hackathon, NIT Raipur (2020)